

6.4 TOBACCO, DRUGS AND ALCOHOL ABUSE (TDA ABUSE)

Recent surveys and statistics show that the use of tobacco, drugs and alcohol has been on the rise especially among the youth. This is really a cause of concern as it could result in many harmful effects. Proper education and guidance would enable the youth to safeguard themselves against these dangerous habits and follow healthy lifestyles. Any addict requires counselling and medical help to get rid of the habit.

6.4.1 Tobacco

Tobacco has been used by human beings for more than 400 years. It contains a large number of chemical substances including **nicotine**, an alkaloid. While buying cigarettes one cannot miss the statutory warning printed on the packet '**Smoking is injurious to health**'.

Mode of abuse: It is smoked or chewed as 'gutkha' or used in the form of 'snuff'.

Effect: Smoking increases the carbon monoxide (CO) level and reduces the oxygen level in the blood. Nicotine stimulates the adrenal gland to release adrenaline and nor-adrenaline into blood. These hormones raise the blood pressure and increase the heart rate. Smoking is associated with bronchitis, emphysema, coronary heart disease, gastric ulcers and increases the incidence of cancers of throat, lungs, urinary bladder etc. Smoking also paves the way to addiction to hard drugs^[1]. Yet, smoking is very prevalent in society, both among the young and old. Tobacco chewing is associated with increased risk of cancer of the oral cavity.

Do you think **passive smoking** is also dangerous? Why?

6.4.2 Drugs

Drugs are the chemical substances used in the treatment, cure and prevention of diseases so as to enhance one's physical or mental well being. For hundreds of years, several plants, fruits and seeds with hallucinogenic properties have been used in folk-medicine, religious ceremonies (such as '**bhang**' on the '**Holi festival**' day) and rituals all over the globe. **When these are taken for a purpose other than the medicinal use or in excess amounts that impair one's physical or psychological functions, it constitutes 'drug abuse'.**

The drugs commonly abused are **opioids**, **cannabinoids** and **coca alkaloids**. Majority of them are obtained from flowering plants but some are obtained from certain fungi.

1. **Opioids:** These are the drugs obtained from opium poppy plant *Papaver somniferum* (vernacular name: '**Nallamandu**'



Fig. 6.14 Opium poppy

^[1] Hard drugs refer to morphine, heroin, cocaine, etc.

mokka'). They bind to specific opioid receptors present in our central nervous system and gastrointestinal tract. Some of the opioids are morphine, heroin, etc.

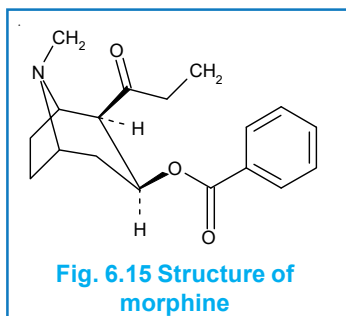


Fig. 6.16 Leaves of Cannabis

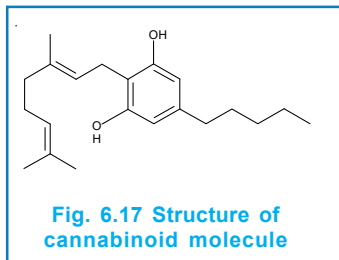


Fig. 6.18 A branch of Datura

- i) **Morphine:** It is extracted from the dried latex of the unripe seed capsule (pod) of poppy plant. It occurs as colourless crystals or a white crystalline powder.

Mode of abuse: Generally it is taken orally or by injection.

Effect: It is a very effective sedative and painkiller. It is very useful in patients who have undergone surgery.

- ii) **Heroin:** It is a white, bitter, odourless and crystalline compound, obtained by the **acetylation of morphine**. Chemically it is **diacetylmorphine**. It is commonly called '**smack**'.

Mode of abuse: Generally it is taken by '**snorting**' and injection.

Effect: Heroin is a depressant and slows down the body functions.

2. **Cannabinoids:** These are a group of chemicals obtained from Indian hemp plant *Cannabis sativa* (vernacular name: **Ganjai mokka**). They interact with cannabinoid receptors present in the brain. The flower tops, leaves and the resin of this plant are used in various combinations to produce marijuana, hashish, charas and ganja. These days, cannabinoids are being abused by even some sports-persons (doping).

Mode of abuse: These are generally taken by **inhalation** and **oral ingestion**.

Effect: Show their effects on cardiovascular system of the body.

3. **Coca alkaloid or Cocaine:** It is a white, crystalline alkaloid that is obtained from the leaves of Coca plant *Erythroxylum coca*, native to South America. It is commonly called '**coke** or **crack**'.

Mode of abuse: It is usually snorted

Effect: It has a potent stimulating action on the central nervous system as it interferes with the transport of the neuro-transmitter '**dopamine**'. Hence it produces a sense of euphoria and increased energy. (NCERT Text) Excessive consumption of crack causes hallucinations.

4. Other well-known plants with hallucinogenic properties are *Atropa belladonna* and *Datura*. Certain drugs like 'Barbiturates (sleeping pills), Amphetamines (cause sleeplessness), Benzodiazepines (tranquilizers), Lysergic acid diethyl amides (LSD) and other similar drugs, normally used as medicines to treat patients with mental illnesses like depression, insomnia, etc.,' are often abused.

6.4.3 Adolescence and TDA abuse

1. Adolescence: It is the time period between the beginning of puberty and the beginning of adulthood. In other words, it is the bridge linking childhood and adulthood. The age between 12-18 years is considered '**adolescence period**'. It is both 'a period and a process' during which a child becomes mature. It is accompanied by several biological and behavioural changes. Thus, adolescence is a very '**vulnerable phase**' of mental and psychological development of an individual.

2. TDA abuse: Curiosity, desire for adventure and excitement, experimentation, are the common causes for the motivation of youngsters towards the use of tobacco, drugs and alcohol. The first use of drugs or alcohol may be out of curiosity or experimentation, but later the person starts using them to escape facing problems. Recently 'stress from the pressure to excel in academics or examinations' has played a significant role in alluring the youngsters to try certain drugs. Television, movies, newspapers and internet also help promoting this wrong perception. Other factors that are associated with tobacco, drug and alcohol abuse among adolescents are unstable or unsupportive family structures and peer pressure.

The consumption of drugs in any form is a non bailable offence. But the seeds of *Opium* are easily available in any provision store. They are treated in such a way that they can never germinate. They are generally used in the preparation of some curries. What are they? **Clue:** Find the Telugu name of poppy seeds.

6.4.4 Addiction and Dependence

The TDA abuse leads to addiction and dependence.

1. Addiction: It is a psychological attachment to certain effects such as euphoria. The most important thing one fails to realise is, the inherent '**addictive nature**' of tobacco, drugs and alcohol. With the repeated use of TDA, the tolerance level of the receptors present in our body increases. Consequently the receptors respond only to higher doses leading to greater intake and addiction. However it should be clearly borne in mind that **use of TDA even once, can be a fore-runner to addiction**. Thus, the addictive potential of tobacco, drugs and alcohol pull the users into a vicious circle leading to their regular use (abuse) from which they may not be able to get out. In the absence of any guidance or counseling, people get addicted and become dependent on them.

2. Dependence: It is the tendency of the body to manifest a characteristic and unpleasant condition (withdrawal syndrome) if the regular dose of drugs or alcohol is abruptly discontinued. The withdrawal syndrome is characterised by anxiety, shakiness (tremors), nausea and sweating which may be relieved when the regular use is resumed again. Dependence leads the patients to ignore all social norms.

6.4.5 Adverse effects of drugs and alcohol abuse

The immediate adverse effects of drugs and alcohol abuse are manifested in the form of reluctant behaviour, vandalism and violence. Excessive doses of drugs may lead to coma and death due to respiratory or heart failure or cerebral haemorrhage. A combination of drugs or their intake along with alcohol generally results in overdosing and even death. **The most common warning signs of drug and alcohol abuse among the youth include ‘drop in academic performance, lack of interest in personal hygiene, depression, fatigue, aggressive behaviour, loss of interest in hobbies, change in sleeping and eating habits, fluctuations in weight, appetite, etc’.**

Those who take drugs intravenously are much more likely to acquire serious infections such as HIV, HBV (Hepatitis-B virus), etc., as the viruses are transferred from one person to another by the sharing of infected needles and syringes. The chronic use of drugs and alcohol damages nervous system and liver. The use of drugs and alcohol during pregnancy is also known to affect the foetus adversely.

Some sports-persons take drugs such as anabolic steroids to enhance their performance. The side-effects of the use of these drugs in females include masculinisation, increased aggressiveness, mood swings, depression, abnormal menstrual cycles, excessive hair growth on the face and body and the enlargement of clitoris. In males it includes acne (pimples), increased aggressiveness, mood swings, depression, reduction in the size of testicles, decreased sperm production, kidney and liver dysfunction, enlargement of breasts, premature baldness and the enlargement of the prostate gland.

6.4.6 Prevention and Control

The age-old adage of **‘Prevention is better than cure’** holds true here also. Some of the measures useful for prevention and control of TDA abuse among the adolescents are:

- i) **Avoid undue parental pressure:** Every child has his/her own choice, capacity and personality. The parents should not force their children to perform beyond their capacity by comparing them with others in studies, games, etc.
- ii) **Responsibility of parents and teachers:** They should look for the danger signs and counsel such students who are likely to get into the ‘trap’.
- iii) **Seeking help from peers:** If peers find some one abusing drugs or alcohol, immediately it should be brought to the notice of their parents or teachers so that they can guide them appropriately.
- iv) **Education and counselling:** Educating and counselling the children to face problems, stress and failures as a part of life.
- v) **Seeking professional and medical help:** A lot of help is available in the form of highly qualified psychologists, psychiatrists and de-addiction and rehabilitation programmers.

GLOSSARY

Abscess: A wound consisting of a localized collection of pus surrounded by inflamed tissue

Abuse: Improper or excessive use.

Alkaloid: Natural bases containing nitrogen, found in plants. e.g. Quinine

Alveoli of lungs: Tiny sacs for holding air in the lungs in which the exchange of gases takes place

Anabolic steroids: Technically known as anabolic-androgen steroids (AAS) or colloquially as 'steroids', are the drugs that mimic the effects of 'testosterone and dihydrotestosterone in the body'. They increase protein synthesis within cells, which results in the buildup of cellular tissue (anabolism), especially in muscles.

Anaemia: Deficiency of red blood cells or haemoglobin

Appetite: A feeling of hunger

Bout: An instance of something lasting for a short period as in bout of fever

Clinical symptoms: Noticeable symptoms of a disease

Cloaca: The cavity at the end of the digestive tract into which the intestinal, genital, and urinary tracts open in the case of vertebrates (only intestinal and genital ducts open in the case of invertebrates)

Constipation: Irregular and infrequent or difficult egestion

Coprophagous: Feeding on faecal matter

Cosmopolitan: Occurring in many parts of the world

Emaciation: Extreme leanness

Epidemic: Spreading of an infectious disease, affecting many people over a wide area

Epidemiology: The branch of medical science dealing with the transmission and control of disease

Euphoria: A temporary feeling of an exaggerated joy, well-being, pride and optimism associated with drugs and alcohol; or an absence of depression

Faeces: Solid indigested product discharged through anus

Filiform: Resembling a thread

Folk-medicine: Treatment given by some aged local people (naatu vaidyam in Telugu)

Giddiness: A feeling that you are about to fall; a reeling sensation

Groin: The junction of the inner part of the thigh with the trunk together with the adjacent region and often including the external genitals

Hallucinations: Illusionary perception; a sensory experience of something that does not exist outside the mind, caused by physical and mental disorders

Hoarseness: Harshness of the throat

Inflammation: A response of body tissues to injury or irritation, characterized by pain, swelling, redness and burning sensation

Instinct: Inborn pattern of behaviour

Intravenous: Direct injection into the vein using a needle and syringe

Lesion: An injury to living tissue

Liver sinusoids: Tiny blood filled spaces in the liver

Masculinisation: Development of features like those of males

Moultings: Periodic shedding of the cuticle or the outer skin in some animals or larval stages

Nasal congestion: Blockage of nose

Neuro-transmitter: A neuro-chemical that transmits nerve impulses across a synapse

Ovoviviparous: Producing eggs that hatch within the body but the juveniles do not draw any nourishment from the mother

Pathogen: Any disease-producing agent

Peer: A person who is of equal standing with another in a group; a friend or a classmate

Phasmids: Caudal sensory organs present in nematodes

Pronucleus: Nuclei of the sperm and ovum before their union to form synkaryon

Quagmire: A soft wet area of low-lying land that sinks underfoot (quag means 'oobi/ oobi nela' in Telugu)

Rituals: The prescribed procedure for conducting religious ceremonies

Sausage shaped: More or less a long-brinjal shaped

Schizogony: A type of multiple fission that occurs in many apicomplexan protozoans during asexual cycle

Snorting: An act of forcible inhalation of drugs

Splenomegaly: An abnormal enlargement of the spleen

Stool: Faecal matter

Synkaryon: Nucleus of the zygote

Thrive: Live/grow vigorously

Tranquilizers: A drug used to reduce stress or tension without reducing mental clarity

Tropical sprue: A chronic disorder that occurs in both children and adults; nutrients are not properly absorbed; symptoms include foul-smelling diarrhoea and emaciation

Ulcer: A circumscribed inflammatory lesion on the skin or an internal mucous membrane resulting in the necrosis of tissue

Vaccine: Immunogen consisting of a suspension of weakened or dead pathogenic cells injected in order to stimulate the production of antibodies in the host body

Vandalism: Deliberate destruction, defacement or damage of public or other people's property

Vermiform: Long, thin and cylindrical or resembling a worm

Vesicular nucleus: Nucleus with a nucleolus and large amounts of euchromatin

Vicious circle: One trouble leads to another that aggravates the first

Vulnerable: Susceptible to attack or temptation

Welfare: Something that aids or promotes

SOME MORE COMMON DISEASES

i) **Bronchitis:** Inflammation of the membranes lining the bronchial tubes

ii) **Chikungunya:** It is a viral disease caused by *Chikungunya virus* (CHIKV) which is a heat sensitive RNA virus
Mode of infection: Inoculation through the bite of *Aedes aegypti* which bites during the day time.

Symptoms: Fever upto 104°F, chills, vomiting, nausea, headache and **severe joint pains** and sometimes it is accompanied with rashes

iii) **Diphtheria:** It is caused by a gram-positive bacillus bacterium *Corynebacterium diphtheriae*. It primarily affects the mucous membranes of the respiratory tract.

Mode of infection: Contamination through airborne respiratory droplets or by direct contact with nasopharyngeal secretions

Symptoms: Sore throat, fever, malaise, hoarseness, difficulty in swallowing or in breathing

iv) **Dysentery:** It is an intestinal inflammation, especially in the colon, caused by *Shigella*, a gram negative bacterium

Mode of infection: Contamination through food & water

Symptoms: Abdominal pain, nausea, vomiting, watery diarrhea which can contain blood, mucus or pus, painful egestion and dehydration

- v) **Emphysema:** an abnormal condition of the lungs marked by decreased respiratory function due to the rupture of alveoli

- vi) **Plague:** Bubonic plague is the most common form of plague. It is caused by a Gram-negative bacterium *Yersinia pestis*

Mode of infection: Inoculation through the bites of an infected flea (*Xenopsylla cheopis*)

Symptoms: Swollen, tender lymph glands, Fever, Chills, Headache and weakness

- vii) **Polio:** It is caused by *Poliomyelitis* virus which is a single-stranded RNA virus

Mode of infection: Contamination through the stool of the infected person.

Symptoms: Inflammation of nerve cells of the brain stem and spinal cord resulting in the impairment of legs.

- viii) **Smallpox:** It is caused by the *Variola* virus.

Mode of infection: Contamination through the tiny drops of an infected person's saliva that comes when the patient talks, coughs or sneezes.

Symptoms: Formation of small pus-filled blisters through out the body, fever, headache, backache, and feeling tired.

- ix) **Tetanus:** It is an infectious disease caused by contamination of wounds from a Gram-positive bacterium *Clostridium tetani* or its spores present in the soil.

Mode of infection: Contamination through soil or dust.

Symptoms: Tetanus results in severe, uncontrollable muscle spasms as in lockjaw.

Muscle spasm: a painful and involuntary muscular contraction.

VERY SHORT ANSWER TYPE QUESTIONS

1. Define parasitism and justify this term.
2. Distinguish between vector and a reservoir host.
3. Distinguish between mechanical vector and biological vector.
4. What is a hyper-parasite? Mention the name of one hyper-parasite.
5. What do you mean by parasitic castration? Give one example.
6. What are the endo-parasitic adaptations observed in *Fasciola hepatica*?
7. Define neoplasia. Give one example.
8. Define the most accurate definition of the term 'health' and write any two factors that affect the health.
9. Distinguish between infectious and non-infectious diseases. Give two examples each.
10. '*Entamoeba histolytica* is an obligatory anaerobe'. Justify.
11. Distinguish between precystic stage and cystic stage of *E. histolytica*.
12. What is the reserve food in the precystic and early cyst stages of *Entamoeba histolytica*?
13. A person is suffering from bowel irregularity, abdominal pain, blood and mucus in stool, etc. Based on these symptoms, name the disease and its causative organism.
14. On the advice of a doctor, a patient has gone to a clinical laboratory for the examination of a sample of faeces. The lab technician, on observing the stool of the patient diagnosed that the patient was suffering from amoebiasis. Write any two

characteristic features based on which the technician came to that conclusion.

15. Define 'asymptomatic cyst passers' with reference to *Entamoeba histolytica*.
16. What are the stages of *Plasmodium vivax* that infect the hepatocytes of man?
17. Define prepatent period. What is its duration in the life cycle of *Plasmodium vivax*?
18. Define incubation period. What is its duration in the life cycle of *Plasmodium vivax*?
19. What are Schuffner's dots? What is their significance?
20. What are haemozoin granules? What is their significance?
21. What is exflagellation and what are the resultant products called?
22. Why is the syngamy found in *Plasmodium* called anisogamy?
23. What is ookinete? Based on the 'sets of chromosomes' how do you describe it?
24. A person is suffering from chills and shivering and high temperature. These symptoms are cyclically followed by profuse sweating and return to normal body temperature. Based on these symptoms, name the disease and its causative organism.
25. Describe the methods of biological control of mosquitoes.
26. The eggs of *Ascaris* are called 'mammillated eggs'. Justify.
27. What is meant by nocturnal periodicity with reference to the life history of a nematode parasite you have studied?

28. Distinguish between lymphadenitis and lymphangitis?
29. 'Elephantiasis is the terminal condition of filariasis'. Justify.
30. In which way does tobacco affect the respiration? Name the alkaloid found in tobacco.
31. Define drug abuse.
32. From which substances 'Smack' and 'Coke' are obtained?
33. 'Many secondary metabolites of plants have medicinal properties. It is their misuse that creates problems'. Justify the statement with an example.
34. Why are cannabinoids and anabolic steroids banned in sports and games?
35. Mention the names of any four drugs which are used as medicines to treat patients with mental illness like depression, insomnia, etc.' that are often abused

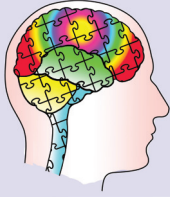
SHORT ANSWER TYPE QUESTIONS

1. What is the need for parasites to develop special adaptations? Mention some special adaptations developed by the parasites.
2. Distinguish between hypertrophy and hyperplasia with an example for each.
3. Describe the structure of a trophozoite of *Entamoeba histolytica*.
4. Explain the life cycle of *Entamoeba histolytica*.
5. Write a short note on the pathogenicity of *Entamoeba histolytica*.
6. Describe the structure of sporozoite of *Plasmodium vivax*.
7. Describe the cycle of Golgi in the life history of *Plasmodium vivax*.
8. Explain the pathogenicity of *Wuchereria bancrofti* in man.

9. Write short notes on typhoid fever and its prophylaxis.
10. Write short notes on pneumonia and its prophylaxis.
11. Write short notes on common cold and its prophylaxis.
12. Write short notes on ringworm and its prophylaxis.
13. What are the adverse effects of tobacco?
14. Write short notes on opioids.
15. Write short notes on cannabinoids.
16. Write short notes on cocaine.
17. Why in adolescence is considered vulnerable phase?
18. Distinguish between addiction and dependence.
19. 'Prevention is better than cure'. Justify with regard to TDA abuse.

LONG ANSWER TYPE QUESTIONS

1. Explain the structure and life cycle of *Entamoeba histolytica* with the help of neat and labelled diagrams.
2. Describe the life cycle of *Plasmodium vivax* in man.
3. Describe the life cycle of *Plasmodium vivax* in mosquito.
4. Describe the life cycle of *Ascaris lumbricoides* with the help of a neat and labelled diagram.
5. Describe the life cycle of *Wuchereria bancrofti*.



For ignited minds

- NOT FOR EVALUATION

1. Why should *Anopheles* only be the vector for *Plasmodium* when there are several other types of mosquitoes in the world ?
2. The epithelium surrounding the wall of the gut is called peritoneum. Inflammation of peritoneum is called peritonitis. From the information provided above, which of the following can possibly cause peritonitis in man? (List: *Plasmodium* , *Wuchereria*, *Entamoeba* and *Ascaris*).
3. If there is swelling in both the legs, it is probably a problem related to heart or kidneys. If there is a swelling in only one leg what should a doctor suspect, in your opinion.
4. Can housefly be both mechanical and biological carrier of pathogens?
5. Hypothetically if the sausage shaped larva (First stage larva) of *Wuchereria bancrofti* is introduced into blood of man , what can be expected as further consequence?
6. Probably Ronald Ross could have missed his Nobel Prize ,had there not been a scientist by the name Patrick

Manson was not born before Ross. If your answer is 'probably, yes' how do you offer an explanation in favour of your view?

7. If human stool has 'live trophozoites' of a parasite, in a sample of stool under examination, how can you identify them as those of *Entamoeba histolytica*? (if you are provided with the right stains and microscope required)
8. In the case of a suspected filarial patient why is the patient admitted in the hospital and a sample of blood is taken from him around midnight.
9. If all the female *Culex* mosquitoes in the world are hypothetically killed will there be filariasis in the world, any more ?
10. If the salivary glands of all female mosquitoes are hypothetically removed, what important effects can be expected? Can you think of at least two important effects?
11. Can you guess the expansion of 'COLD'?

Common obstructive lung disease.